Two routes from a zero-cancellation condition to the Riemann hypothesis

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Abstract

We study a zero-cancellation condition for the Bernstein rows associated with the actual normalized logarithmic-derivative moments of the Riemann ξ-function. The condition, denoted H_{00}, asserts admissibility of the cutoff-zero, coefficient-zero cancellation budget. We present two constructions of the implication $H_{00} \Rightarrow$ RH. The first passes from constant row variation to an existing actual spectral criterion. The second constructs a Hausdorff probability measure, identifies its resolvent with the actual logarithmic derivative near the origin, and continues an analytic defect to exclude nonreal zeros. A retained converse gives $H_{00} \Longleftrightarrow$ RH for each construction. The accompanying Lean development exports both implications with no additional mathematical hypothesis. The condition H_{00} remains a universally quantified assertion: neither its unconditional validity nor a terminating finite certificate checker is established here.

Keywords. Riemann hypothesis; Hausdorff moments; Bernstein weights; logarithmic derivative; positive measure; formal proof.

1 Introduction

Moment positivity provides a natural meeting point between finite algebraic identities and the analytic location of zeros. In the present setting, the finite objects are Bernstein rows formed from a fixed, actual sequence of normalized logarithmic-derivative moments. Their total variation is at least one. Equality at every order forces all row weights to be nonnegative and is equivalent to the vanishing of a nonnegative cancellation quantity.

This note records two ways to pass from that cancellation condition to the Riemann hypothesis. They share the actual moment sequence and the finite positivity argument, but use different forward analytic constructions. The first applies an already formalized spectral characterization of subexponential row variation. The second constructs a positive measure directly from the Bernstein rows and uses its resolvent to control the logarithmic derivative.

The distinction between a finite description and a finite verification is central. The candidate budget consists of two natural numbers, namely $(0, 0)$, but its admissibility quantifies over every row index. The equivalence proved here therefore does not supply an accepted finite certificate of RH. In particular, no proof of H_{00} is claimed.

The mathematical development below is accompanied by the Lean sources in [1]. We explicitly identify the previously formalized actual-function and spectral results used as inputs. The note does not reprove the full analytic infrastructure of that source development. Section 8 distinguishes the mathematical statements, reported native verification, and the separately configured GitHub CI target.

2 Actual analytic data

Let ξ be the actual entire Riemann ξ-function used in the accompanying development. Away from $s = 0, 1$, its normalization is

$$\xi(s) = \frac{s(s-1)}{2}\,\Lambda(s), \tag{2.1}$$

where Λ denotes the completed Riemann zeta function. More exactly, with Mathlib's entire pole-subtracted auxiliary function Λ_0 (`completedRiemannZeta`$_0$), the formal definition is

$$\xi(s) = \frac{1 + s(s-1)\Lambda_0(s)}{2}.$$

This specifies the values at the removable singularities as well. We use the retained equivalence between Mathlib's literal `RiemannHypothesis` and the assertion that every zero of this ξ has real part $1/2$.

The reflection identity $\xi(1-s) = \xi(s)$ makes $w \mapsto \xi(1/2 + w)$ even. The actual function

$$\Theta(z) = \xi\big(\tfrac{1}{2} + \sqrt{z}\big)$$

is proved entire in the source, despite the square root used in its definition, and satisfies

$$\Theta(w^2) = \xi\big(\tfrac{1}{2} + w\big) \qquad (w \in \mathbb{C}). \tag{2.2}$$

The following actual-function facts are retained from that development:

(i) $\Theta(0) \neq 0$, and Θ is entire;

(ii) the zero set of Θ is invariant under complex conjugation;

(iii) if $\xi(s) = 0$, then $|\operatorname{Im} s| > 1$;

(iv) the moments defined below are real before taking real parts, and their normalizing constant is strictly positive.

These are supplied lemmas about the actual functions, not additional hypotheses of the final H_{00} implications. Their source locations are listed in Appendix A. Only the nonvanishing of $\operatorname{Im} s$, rather than the numerical strength of (iii), is needed in the final return from Θ-zeros to ξ-zeros.

On a neighborhood of zero define

$$R(z) = \frac{\Theta'(z)}{\Theta(z)}, \qquad m_n = \operatorname{Re}\!\left(\frac{(-1)^n}{n!} R^{(n)}(0)\right), \qquad \kappa = m_0 > 0, \qquad u_n = \frac{m_n}{\kappa^{n+1}}. \tag{2.3}$$

Here $R^{(0)} = R$. In particular $u_0 = 1$. The retained reality result turns (2.3) into the complex derivative identity

$$R^{(n)}(0) = (-1)^n n!\,\kappa^{n+1} u_n. \tag{2.4}$$

Throughout the paper, unless a general real sequence is explicitly introduced, u denotes this actual normalized sequence.

3 Bernstein cancellation and the main statement

For a real sequence u, define its Bernstein weights by

$$w_{N,j} = \binom{N}{j} \sum_{r=0}^{N-j} (-1)^r \binom{N-j}{r} u_{j+r} \qquad (0 \le j \le N), \qquad V_N = \sum_{j=0}^{N} |w_{N,j}|. \tag{3.1}$$

The elementary row-mass and coarsening identities are

$$\sum_{j=0}^{N} w_{N,j} = u_0, \tag{3.2}$$

$$(n+1)w_{n,j} = (n+1-j)w_{n+1,j} + (j+1)w_{n+1,j+1} \quad (0 \le j \le n). \tag{3.3}$$

For $0 \le j \le n$ put $a_{n,j} = (n+1-j)w_{n+1,j}$ and $b_{n,j} = (j+1)w_{n+1,j+1}$. Define the row cancellation by

$$C_n = \sum_{j=0}^{n} (|a_{n,j}| + |b_{n,j}| - |a_{n,j} + b_{n,j}|). \tag{3.4}$$

The triangle inequality gives $C_n \ge 0$. Summing the weighted terms and using (3.3) yields

$$C_n = (n+1)(V_{n+1} - V_n). \tag{3.5}$$

Definition 3.1 (The semantic zero budget)**.** A cancellation budget is a pair $W = (N_0, b) \in \mathbb{N}^2$. For the actual sequence it is *admissible* if

$$\forall n \ge N_0, \qquad C_n \le bV_n.$$

The condition H_{00} is admissibility of $W = (0, 0)$.

Proposition 3.2. *For the actual normalized sequence, the following are equivalent:*

(i) H_{00}*;*

(ii) $C_n = 0$ *for every* $n \ge 0$*;*

(iii) $V_N = 1$ *for every* $N \ge 0$*;*

(iv) $w_{N,j} \ge 0$ *for every* $0 \le j \le N$*.*

Proof. Since $C_n \ge 0$, admissibility of $(0, 0)$ is equivalent to $C_n = 0$ at every order. Equation (3.5) then says $V_{n+1} = V_n$. Since $V_0 = |u_0| = 1$, induction gives (iii). Conversely, (iii) forces the increment to vanish and hence gives (ii).

By (3.2), each row has mass one. Therefore

$$V_N - 1 = \sum_{j=0}^{N} (|w_{N,j}| - w_{N,j}).$$

Every summand is nonnegative, and it vanishes precisely when the corresponding weight is nonnegative. This proves the equivalence of (iii) and (iv). □

Theorem 3.3 (Two forward constructions and a common converse)**.** *For the actual sequence* (2.3)*,*

$$H_{00} \quad \Longleftrightarrow \quad \text{RH}.$$

The forward implication admits both the spectral construction of Section 4 and the measure–resolvent construction of Sections 5–7. Each construction has H_{00} *as its only mathematical hypothesis. The two equivalences use the same retained converse.*

The proof is completed below. The theorem is an equivalence of mathematical assertions. It does not assert that either assertion holds unconditionally.

4 The spectral route and the converse

Call a nonnegative sequence V *subexponential* if

$$\forall \varepsilon > 0 \ \exists A > 0 \ \forall n \geq 0, \qquad V_n \leq A e^{\varepsilon n}. \tag{4.1}$$

The earlier actual spectral development supplies the following characterization for the specific V in (3.1):

$$V \text{ is subexponential} \quad \Longleftrightarrow \quad \text{RH}. \tag{4.2}$$

This characterization is supplied by the actual spectral theorem in [1]; the exact declaration is listed in Appendix A. Its analytic inputs are constructed in that development rather than assumed at the present endpoints.

More precisely, the canonical actual divisor gives complex nodes $(q_i)_{i \in J}$, with zero multiplicities retained, satisfying

$$q_i \neq 0, \qquad \sum_i |q_i| < \infty, \qquad u_n = \sum_i q_i^{n+1}, \qquad \alpha_i = -\frac{1}{\kappa q_i}. \tag{4.3}$$

Every α_i is a zero of Θ, and every zero occurs in this family. Put

$$\rho = \sup\bigl(\{1\} \cup \{|q_i| + |1 - q_i| : i \in J\}\bigr).$$

The retained extraction theorem, applied to the constructed calibration, proves the ordinary limit

$$\lim_{N \to \infty} V_N^{1/N} = \rho. \tag{4.4}$$

Its proof separates a finite dominant exponential part from a tail with strictly smaller growth; the full extraction argument is in the accompanying source. Since $|q| + |1 - q| \geq 1$, with equality precisely for $q \in [0, 1]$, the case $\rho = 1$ is equivalent to all nodes lying in that segment. The actual calibration identifies this case with RH and yields (4.2). We use that already constructed spectral result here, without reproducing its full proof.

Proof of the first forward implication. Assume H_{00}. Proposition 3.2 gives $V_n = 1$ for every n. For every $\varepsilon > 0$, take $A = 1$ in (4.1), since $1 \leq e^{\varepsilon n}$ for $n \geq 0$. The forward direction of (4.2) proves RH. □

For the common converse, the actual calibration under RH has every $q_i \in [0, 1]$. Substitution into the finite Bernstein formula gives

$$w_{N,j} = \binom{N}{j} \sum_i q_i^{j+1} (1 - q_i)^{N-j} \geq 0.$$

The interchange with the finite sum is justified by the retained absolute summability. Proposition 3.2 now gives H_{00}. This is the source's retained RH-to-zero-budget construction, exposed to both routes as `H00_of_nativeRH`, and supplies the reverse implication of Theorem 3.3. The remainder of the paper gives a different forward proof. It uses Proposition 3.2 but does not invoke (4.2) or the first forward RH endpoint.

5 From positive Bernstein rows to a probability measure

Write $I = [0,1]$. For the moment, let $u = (u_n)_{n\geq 0}$ be an arbitrary real sequence and put

$$w^u_{N,j} = \binom{N}{j} \sum_{r=0}^{N-j} (-1)^r \binom{N-j}{r} u_{j+r}, \qquad 0 \leq j \leq N.$$

Define $q_{N,j} = j/N$ when $N > 0$, and define $q_{0,0} = 0$. The finite Bernstein average is

$$A^u_N(f) = \sum_{j=0}^{N} w^u_{N,j} f(q_{N,j}). \tag{5.1}$$

In particular, $A^u_0(f) = u_0 f(0)$. This degree-zero convention has no effect on limits as $N \to \infty$.

Lemma 5.1 (Finite moment convergence)**.** *For every real sequence u and every integer $k \geq 0$,*

$$\lim_{N\to\infty} A^u_N(q \mapsto q^k) = u_k.$$

No positivity or representing-measure assumption is required.

Proof. Let $u^+_j = u_{j+1}$. The elementary binomial identity $(j+1)\binom{N+1}{j+1} = (N+1)\binom{N}{j}$ gives the exact shifted-row relation

$$(j+1) w^u_{N+1,j+1} = (N+1) w^{u^+}_{N,j}.$$

The term at $j = 0$ contributes zero to a positive-degree monomial. Hence

$$A^u_{N+1}(q^{k+1}) = \sum_{j=0}^{N} w^{u^+}_{N,j} \left(\frac{j+1}{N+1}\right)^k.$$

For $N \geq 1$, expand

$$\frac{j+1}{N+1} = \frac{N}{N+1}\frac{j}{N} + \frac{1}{N+1}$$

by the binomial theorem. Interchanging two finite sums yields

$$A^u_{N+1}(q^{k+1}) = \sum_{i=0}^{k} T_{N,k,i} A^{u^+}_N(q^i), \qquad T_{N,k,i} = \binom{k}{i} \left(\frac{N}{N+1}\right)^i \left(\frac{1}{N+1}\right)^{k-i}. \tag{5.2}$$

For $0 \leq i \leq k$, $T_{N,k,i} \to \mathbf{1}_{i=k}$. The row-mass identity $\sum_j w^u_{N,j} = u_0$ proves the degree-zero case. Strong induction on the degree, applied to the shifted sequence u^+ in (5.2), now gives the required limit. □

Assume henceforth that $u_0 = 1$ and that every Bernstein weight is nonnegative. The row-mass identity makes each A^u_N a positive, unital linear functional on $C(I, \mathbb{R})$, and

$$|A^u_N(f)| \leq \|f\|_\infty. \tag{5.3}$$

Only values on the finite grid enter these averages.

Proposition 5.2 (A constructed Hausdorff measure)**.** *If $u_0 = 1$ and $w^u_{N,j} \geq 0$ for all $0 \leq j \leq N$, there is a Borel probability measure σ on I such that*

$$\int_I q^k \, d\sigma(q) = u_k \qquad (k \geq 0). \tag{5.4}$$

Proof. Fix one free ultrafilter $\mathcal{U}$ on $\mathbb{N}$ refining the cofinite filter. By (5.3), the sequence $A_N^u(f)$ lies in the compact interval $[-\|f\|_\infty, \|f\|_\infty]$. Its $\mathcal{U}$-limit therefore exists for every $f \in C(I, \mathbb{R})$. Set

$$L(f) = \lim_{N \to \mathcal{U}} A_N^u(f).$$

The same ultrafilter is used for every f. Uniqueness of limits and the finite linear identities imply linearity of L. Order passes to the limit, so L is positive; moreover, $L(1) = 1$ and $|L(f)| \le \|f\|_\infty$. The Riesz–Markov–Kakutani representation theorem on the compact space I supplies a positive measure σ with $L(f) = \int_I f \, d\sigma$. Taking $f = 1$ gives $\sigma(I) = 1$. Finally, ordinary convergence in Lemma 5.1 implies convergence to the same value along $\mathcal{U}$, giving (5.4). □

For the actual normalized moment sequence, $u_0 = 1$ is a retained unconditional identity and H_{00} implies all the required Bernstein signs. Thus Proposition 5.2 constructs the measure from H_{00}; measure existence is not an extra premise. This classical construction uses the fixed free ultrafilter and the Riesz representation theorem, and should not be interpreted as a terminating finite checker.

6 The analytic resolvent and its actual germ

Let $\kappa > 0$ be the actual normalizing constant and let σ be the probability measure just constructed. Define

$$S(z) = \kappa \int_I \frac{d\sigma(q)}{1 + \kappa q z}, \qquad D_\kappa = \{z \in \mathbb{C} : \ 1 + \kappa q z \neq 0 \text{ for all } q \in I\}. \tag{6.1}$$

Certainly $0 \in D_\kappa$. Also $\mathbb{H} = \{z : \operatorname{Im} z > 0\}$ is contained in D_κ: for $q > 0$ the denominator has positive imaginary part, while at $q = 0$ it is 1.

Lemma 6.1 (Kernel differentiation)**.** *The function S is analytic on D_κ, and for every $n \ge 0$,*

$$S^{(n)}(z) = (-1)^n n! \, \kappa^{n+1} \int_I \frac{q^n}{(1 + \kappa q z)^{n+1}} \, d\sigma(q) \qquad (z \in D_\kappa). \tag{6.2}$$

Proof. Fix $z_0 \in D_\kappa$. Compactness of I gives $a = \min_{q \in I} |1 + \kappa q z_0| > 0$. If $|z - z_0| < a/(2\kappa)$, the reverse triangle inequality gives $|1 + \kappa q z| \ge a/2$ uniformly in q. This also proves that D_κ is open. For

$$K_n(z, q) = \frac{q^n}{(1 + \kappa q z)^{n+1}},$$

one has $\partial_z K_n = -(n+1)\kappa K_{n+1}$ and, on the indicated neighborhood, $|K_n(z, q)| \le (a/2)^{-n-1}$. The derivative is likewise bounded by a constant independent of q. Since σ is finite, these constants are integrable majorants. Differentiation under the integral is therefore valid. It proves complex differentiability locally throughout D_κ, hence analyticity, and induction yields (6.2). □

For $z \in \mathbb{H}$, the elementary kernel identity gives

$$\operatorname{Im} S(z) = -\kappa^2 \operatorname{Im} z \int_I \frac{q}{|1 + \kappa q z|^2} \, d\sigma(q) \le 0. \tag{6.3}$$

Positivity of the measure is essential here.

Let $R = \Theta'/\Theta$ be the actual logarithmic derivative. The retained nonvanishing theorem $\Theta(0) \neq 0$ makes R analytic near zero. The definition of the actual normalized moments, together with the retained reality theorem for the complex moments, gives

$$R^{(n)}(0) = (-1)^n n!\, \kappa^{n+1} u_n.$$

On the other hand, evaluating (6.2) at zero and using (5.4) gives the same expression for $S^{(n)}(0)$. Both functions are analytic there; their convergent Taylor expansions consequently agree on a neighborhood of zero:

$$R(z) = S(z) \quad \text{for all sufficiently small } |z|. \tag{6.4}$$

This is equality of analytic germs, not merely equality of formal coefficient lists. No global zero-free assertion about Θ was used. The germ (6.4), analyticity on $\mathbb{H}$, and the sign (6.3) are the inputs to the analytic-defect continuation and pole-exclusion argument.

7 Continuation and exclusion of nonreal zeros

7.1 Continuation of the analytic defect

Write $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$. The measure construction provides a function S holomorphic on $\mathbb{H}$ and on a neighborhood of the origin, with

$$\operatorname{Im} S(z) \leq 0 \quad (z \in \mathbb{H}), \qquad S(z) = R(z) \quad \text{for all sufficiently small } z,$$

where $R = \Theta'/\Theta$ is used only at points where $\Theta \neq 0$. The latter germ is well-defined because the retained actual-function development proves $\Theta(0) \neq 0$. We now extend the germ in a form that is holomorphic even at possible zeros of Θ.

Lemma 7.1 (Analytic-defect continuation)**.** *Let F be entire and let S be holomorphic on $\mathbb{H}$. If there exists $\varepsilon > 0$ such that $F' = SF$ on $\mathbb{H} \cap B(0, \varepsilon)$, then $F' = SF$ throughout $\mathbb{H}$.*

Proof. The function $D = F' - SF$ is holomorphic on $\mathbb{H}$. Choose $\varepsilon > 0$ as in the hypothesis, so $D = 0$ on $\mathbb{H} \cap B(0, \varepsilon)$. Although 0 lies on the boundary of $\mathbb{H}$, the point $z_* = i\varepsilon/2$ lies in both $\mathbb{H}$ and $B(0, \varepsilon)$. Hence D vanishes on a neighborhood of z_* contained in $\mathbb{H}$. The upper half-plane is convex and therefore connected. The identity theorem gives $D \equiv 0$ on $\mathbb{H}$. □

Apply the lemma with $F = \Theta$. Continuity and $\Theta(0) \neq 0$ give an origin neighborhood on which Θ does not vanish. The equality of germs $R = S$ therefore implies $\Theta' = S\Theta$ there, and analytic-defect continuation yields

$$\Theta'(z) = S(z)\Theta(z) \qquad (z \in \mathbb{H}). \tag{7.1}$$

In particular,

$$\frac{\Theta'(z)}{\Theta(z)} = S(z), \qquad \operatorname{Im} \frac{\Theta'(z)}{\Theta(z)} \leq 0 \quad (z \in \mathbb{H},\ \Theta(z) \neq 0). \tag{7.2}$$

The continuation argument requires no assertion that Θ is already zero-free: it is applied to D, not to a quotient across its possible poles.

7.2 The sign condition excludes nonreal zeros

Lemma 7.2 (Upper-half-plane pole obstruction)**.** *Let F be a nonzero entire function. Suppose*

$$\operatorname{Im} \frac{F'(z)}{F(z)} \le 0 \qquad (z \in \mathbb{H},\ F(z) \neq 0).$$

Then F has no zero in $\mathbb{H}$.

Proof. Assume that $F(\rho) = 0$ with $\operatorname{Im} \rho > 0$. Since F is entire and not identically zero, its zero at ρ has a finite multiplicity $m \ge 1$. On a neighborhood of ρ we have

$$F(z) = (z-\rho)^m g(z), \qquad g(\rho) \neq 0,$$

with g holomorphic. Shrinking the neighborhood makes g nonvanishing there. Thus, for $z \neq \rho$ in that neighborhood,

$$\frac{F'(z)}{F(z)} = \frac{m}{z-\rho} + \frac{g'(z)}{g(z)}. \tag{7.3}$$

Set $B = g'/g$. Continuity provides $K > 0$ and $\varepsilon_0 > 0$, with $\varepsilon_0 < \operatorname{Im} \rho$, such that

$$\bigl|\operatorname{Im} B(\rho - i\varepsilon)\bigr| \le K \qquad (0 < \varepsilon < \varepsilon_0).$$

The same choice ensures that $F(\rho - i\varepsilon) \neq 0$. Choose

$$0 < \varepsilon < \min\{\varepsilon_0, m/K\}.$$

Then $z_\varepsilon = \rho - i\varepsilon$ remains in $\mathbb{H}$, while (7.3) gives

$$\operatorname{Im} \frac{F'(z_\varepsilon)}{F(z_\varepsilon)} = \frac{m}{\varepsilon} + \operatorname{Im} B(z_\varepsilon) \ge \frac{m}{\varepsilon} - K > 0.$$

This contradicts the assumed sign condition. □

For the actual function Θ, nontriviality follows from $\Theta(0) \neq 0$. Applying the lemma to (7.2) excludes all upper-half-plane zeros. The retained conjugate-zero symmetry

$$\Theta(z) = 0 \quad \Longrightarrow \quad \Theta(\overline{z}) = 0$$

then excludes lower-half-plane zeros as well. Consequently every zero of Θ is real. Only symmetry of the zero set is needed at this step; a separate global Schwarz-reflection hypothesis is not added to the endpoint.

7.3 Return to the Riemann hypothesis

The actual-function development supplies the centered-square identity

$$\Theta\bigl((s-\tfrac{1}{2})^2\bigr) = \xi(s)$$

and the unconditional bound $|\operatorname{Im} s| > 1$ whenever $\xi(s) = 0$. Let $s = \sigma + it$ be such a zero. The centered-square identity and the reality of every Θ-zero imply

$$0 = \operatorname{Im}\bigl((s-\tfrac{1}{2})^2\bigr) = 2(\sigma - \tfrac{1}{2})t.$$

The height bound implies $t \neq 0$, so $\sigma = 1/2$. The retained equivalence between this statement about the actual ξ-zeros and Mathlib's `RiemannHypothesis` completes the second implication.

All analytic hypotheses used above are supplied for the actual objects by the preceding construction and the retained actual-function lemmas. In particular, no local-pole existence assumption, global zero-free assumption, or additional moment-identification premise remains in the final theorem. The forward proof does not invoke the first route's RH conclusion, although the two developments share the definition of H_{00} and the finite positivity lemmas. Both equivalences use the same retained converse RH $\Rightarrow H_{00}$.

Completion of Theorem 3.3. Starting from H_{00}, Proposition 3.2 supplies positive normalized Bernstein rows. Proposition 5.2 constructs their probability measure; Lemma 6.1 and equation (6.4) identify the actual resolvent germ. The continuation and pole arguments above give RH. Together with the spectral forward proof and the common converse in Section 4, this proves the claimed equivalences. □

8 Formalization and remaining obligations

The accompanying development uses Lean 4 [2] and Mathlib [3]. Its two forward endpoint types are exactly $H00 \to$ `RiemannHypothesis`; both equivalence declarations have type $H00 \leftrightarrow$ `RiemannHypothesis`. Appendix A records their precise names and the principal source correspondences.

The forward constructions have shared foundations. In particular, the second route imports the definition of $H00$ and the direct finite positivity lemmas from the first route's module. Its forward proof does not call the first route's RH conclusion. The assertion of two routes refers to these two forward mathematical constructions, not to disjoint import trees.

8.1 Verification record

The frozen source package is version 0.2.3. The supplied native Windows Lean 4.33.0 console excerpt reports `PROJECT_BUILD=PASS` and `RUN=PASS`, with three compiled tasks and 115 reused tasks. The preserved plan contains 117 project modules and a final audit source, hence 118 Lean files. The final audit selects 641 declarations, including all four endpoints.

This native result is recorded as a *reported* compiler-and-audit pass: the original complete native logs and the individual axiom lists were not independently supplied with that checkpoint. Inspection of the preserved audit plan and runner confirms the intended coverage and the allowed axiom set

$$\{\texttt{propext}, \texttt{Classical.choice}, \texttt{Quot.sound}\}.$$

The source is preserved without changing its theorem statements or proof bodies. Its historical preparation comments predate the later native PASS record.

GitHub CI is a separate verification target. It pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

and preserves the transitive dependency lock. The native Mathlib revision was not supplied, so equality of native and CI dependency revisions is not assumed. The workflow verifies source hashes, builds the preserved modules, checks the exact endpoint types, and reruns the complete axiom audit. Missing or duplicated declarations and axioms outside the stated set cause failure.

On 28 October 2026, GitHub run 40712534224 completed successfully at source revision `c6b95fd9e3d792270026da083bddddbc72bacb22`. The job result and full log were independently read: the project build, all four endpoint type checks, and the complete

641-declaration axiom audit passed. Each of the four principal endpoints reported exactly `propext`, `Classical.choice`, and `Quot.sound`. The final source-identity check also passed for all 118 preserved Lean files. The run and retained evidence are linked in [4]. This observed CI result is separate from the earlier reported native PASS. No local Lean or Lake execution was used to prepare this manuscript.

8.2 What is not proved

The condition H_{00} is still open as an unconditional assertion about the actual sequence. Establishing finitely many rows does not prove its universal quantifier. A separate propagation theorem or a sound terminating certificate checker would be required to discharge it by finite data.

In particular, for a proposed type of finite raw witnesses and a terminating Boolean checker, the desired statement

$$(\exists W : \text{FiniteWitness},\ \text{check}(W) = \text{true}) \quad \Longleftrightarrow \quad \text{RH}$$

has not been obtained from the constructions in this note. Soundness must connect acceptance to the actual mathematical condition, and completeness is needed for the reverse existence implication. No accepted finite witness is exhibited here.

The second route presented above is the measure–resolvent variant. The original longer Widder/angular-PDE construction is a distinct unfinished formalization. The analytic germ and continuation arguments proved here do not certify that separate chain.

A Formal statements and source correspondence

All four principal declarations belong to `RH_HilbertLimit.TwoRoutes`. Their short names and types are:

```
nativeRH_of_H00_route_one :
  H00 → _root_.RiemannHypothesis
H00_iff_nativeRH_route_one :
  H00 ↔ _root_.RiemannHypothesis
nativeRH_of_H00_route_two :
  H00 → _root_.RiemannHypothesis
H00_iff_nativeRH_route_two :
  H00 ↔ _root_.RiemannHypothesis
```

The defining equation is

```
H00 := zeroBudgetCandidate.Admissible.
```

The spectral characterization is

```
canonical_actual_subexponential_iff_nativeRH,
```

and the ordinary root-limit theorem is

```
canonical_actual_root_tendsto_constructed.
```

Both are in `RH_HilbertLimit/AnalyticClosure.lean`. The repository paths below refer to the fixed source revision in [1]. All paths beginning with `TwoRoutes/` are relative to `RH_HilbertLimit/`.

Mathematical component	Source modules
Actual ξ, centered-square identity, entire Θ	`AWPIRH/XiCore.lean`; `AWPIRH/ThetaEntire.lean`
Conjugate zeros and nonzero zero heights	`AngularMaster/ZeroGeometry.lean`; `AngularMaster/ZeroFreeSeed.lean`
Actual moments and literal RH binding	`RH_HilbertLimit/ActualXi.lean`
Cancellation condition and spectral endpoint	`RH_HilbertLimit/ActualCancellationWitness.lean`; `RH_HilbertLimit/ZeroBudgetCandidate.lean`; `TwoRoutes/RouteOne.lean`
Finite monomial convergence	`TwoRoutes/BernsteinMomentLimit.lean`
Positive functional and exact Hausdorff moments	`TwoRoutes/HausdorffMeasure.lean`
Integral differentiation, sign, actual germ	`TwoRoutes/ResolventAnalytic.lean`; `TwoRoutes/MeasureResolvent.lean`
Analytic-defect continuation and actual pole closure	`TwoRoutes/ActualResolventContinuation.lean`; `TwoRoutes/ActualHerglotzClosure.lean`; `TwoRoutes/RouteTwo.lean`
Full declaration audit	`TwoRoutesAudit.lean`; `audit-targets.tsv`

The frozen chronological project anchor is `ZETA_HYPERBOLIC_ANCHOR_5`. The native package `RH_H00_TWO_ROUTES_LEAN_v0_2_3_construction.zip` has SHA-256:

`578cc8ad072c79dc7838fcf65a27b9eed`
`707fcc98e0fdf6a4b3601aa20e8c51a`

The line break in this display is typographical; the digest is the concatenation of both lines. The native checkpoint is retained as `evidence/NATIVE_PASS_CHECKPOINT.json`.

References

[1] Hilber-Vector-Resolvent-Theory project, *Two-route Lean source development*, version 0.2.3. Repository revision `c6b95fd9e3d792270026da083bddddbc72bacb22`, 2026. `https://github.com/randogelist/Hilber-Vector-Resolvent-Theory/tree/c6b95fd9e3d792270026da083bddddbc72bacb22`.

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[4] Hilber-Vector-Resolvent-Theory project, *GitHub Actions: Lean build and axiom audit*, run 40712534224, source revision as in [1]. `https://github.com/randogelist/Hilber-Vector-Resolvent-Theory/actions/runs/40712534224`.